As a preliminary note

In Abaqus, Rayleigh damping is applied at material level, and it can be limited by the material type used,

1) Abaqus built-in material – both stiffness-proportional and mass-proportional damping are supported;

2) Abaqus user-defined material (UMAT) – only mass-proportional damping is supported.

In this case, we need to additionally implement stiffness-proportional damping in the UVC subroutine. This report aims to show how I modified the UVC subroutine, and used a SDF system for validation.

As a preliminary note

As discussed, I need to run an SDF system using the Abaqus built-in VC model with stiffness-proportional damping for comparison purpose. The original validation scheme is as follows,

1. **Elastic** SDF system dynamic analysis – compare between 1) VC model with built-in stiffness-proportional damping, and 2) modified UVC model;

2. **Inelastic** SDF system dynamic analysis – compare between 1) VC model with built-in stiffness-proportional damping, and 2) modified UVC model.

However, I later found that, for beam-column elements, the VC material is not supported. If we use built-in material, only kinematic or isotropic hardening can be assigned to these elements – not combined hardening. <u>If we want to assign combined hardening to these elements, it has to be via UMAT like UVC</u>. Nikos mentioned he had also encountered this limitation before. Therefore, I made the following adjustment to the validation plan:

1. **Elastic** SDF system dynamic analysis – compare between 1) **built-in elastic material** with built-in stiffness-proportional damping, and 2) modified UVC model;

2. **Inelastic** SDF system dynamic analysis – compare between 1) **original UVC model with built-in mass-proportional damping**, and 2) modified UVC model. The two models here should achieve the same damping ratio, so the comparison can still make sense.

Topics

Based on the previous discussions, the topics of this report includes,

1. The configuration and eigenvalue analysis of the SDF system investigated;

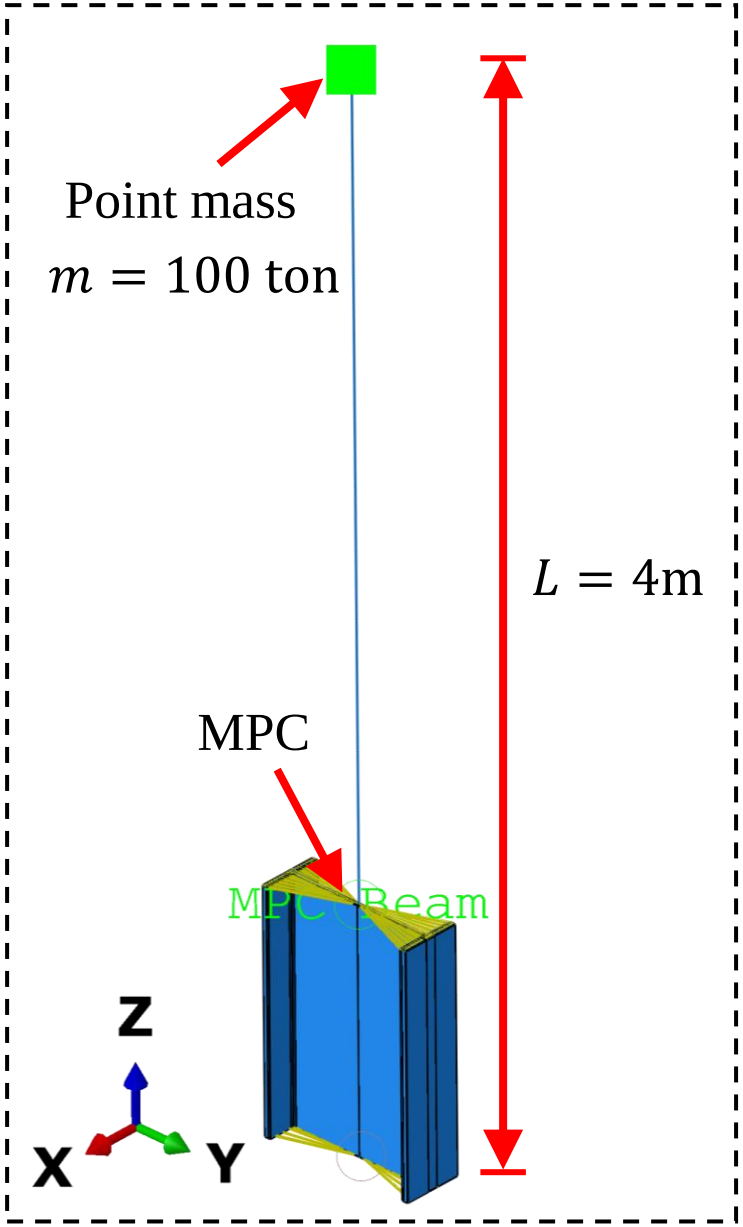
2. Modifications of the UVC model for considering stiffness-proportional damping;

3. Elastic SDF system dynamic analysis – comparing built-in elastic material and modified UVC model;

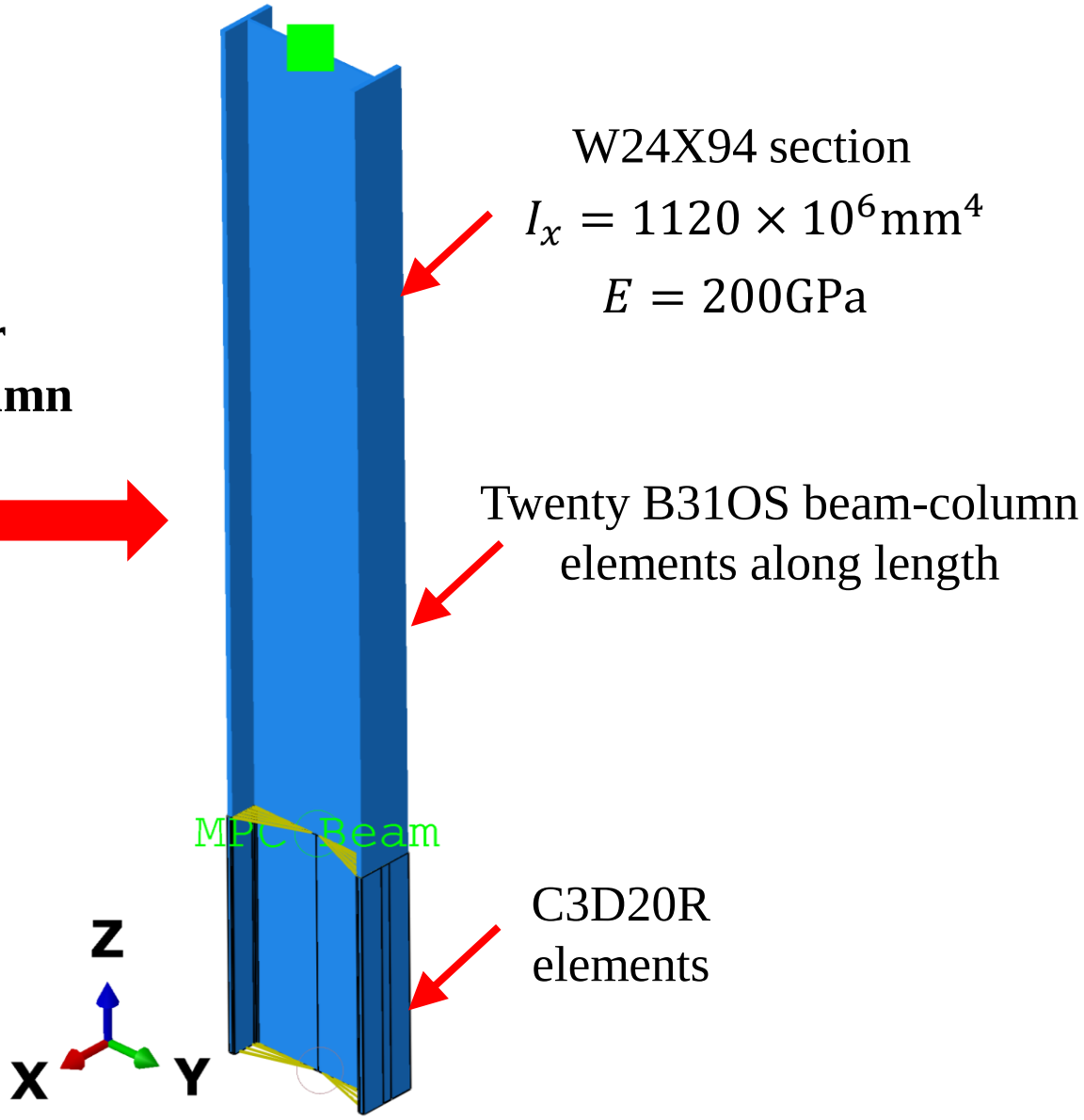
4. Inelastic SDF system dynamic analysis – comparing the original UVC model with mass-proportional damping and modified UVC model.

1. SDF system configuration and eigenvalue analysis

1. SDF system configuration – model description

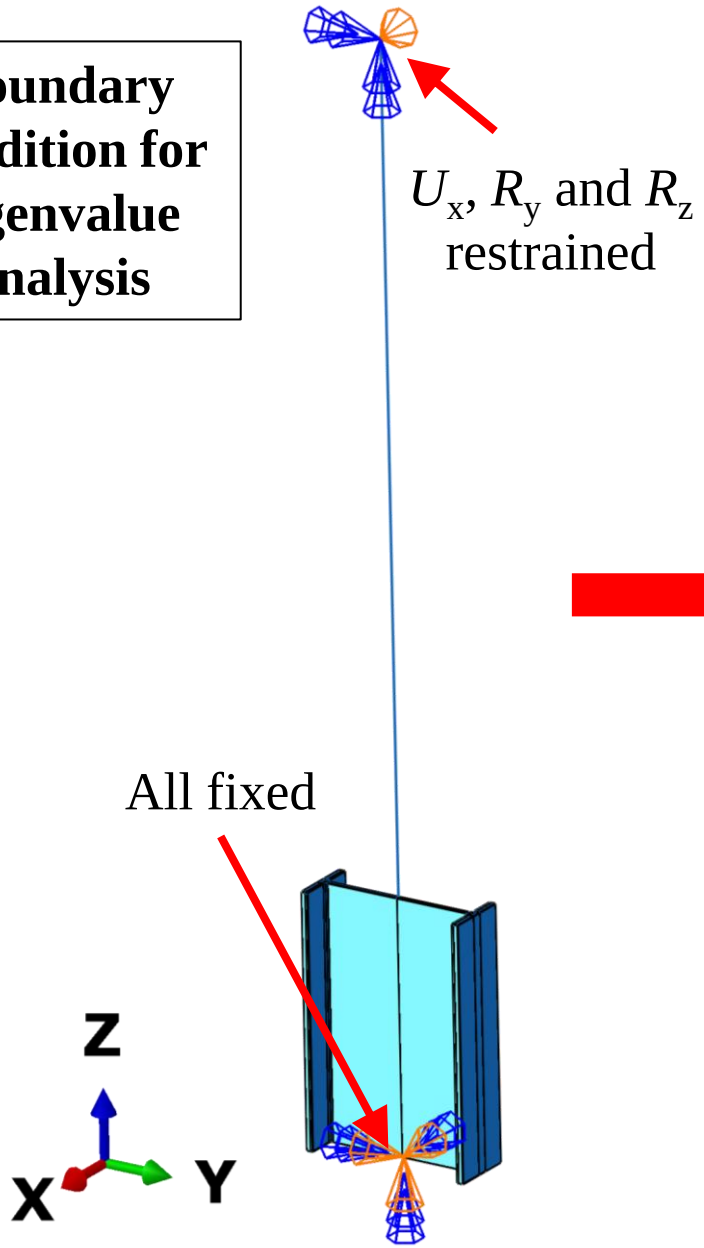


Render beam-column profile



1. SDF system configuration – eigenvalue analysis

Boundary condition for eigenvalue analysis

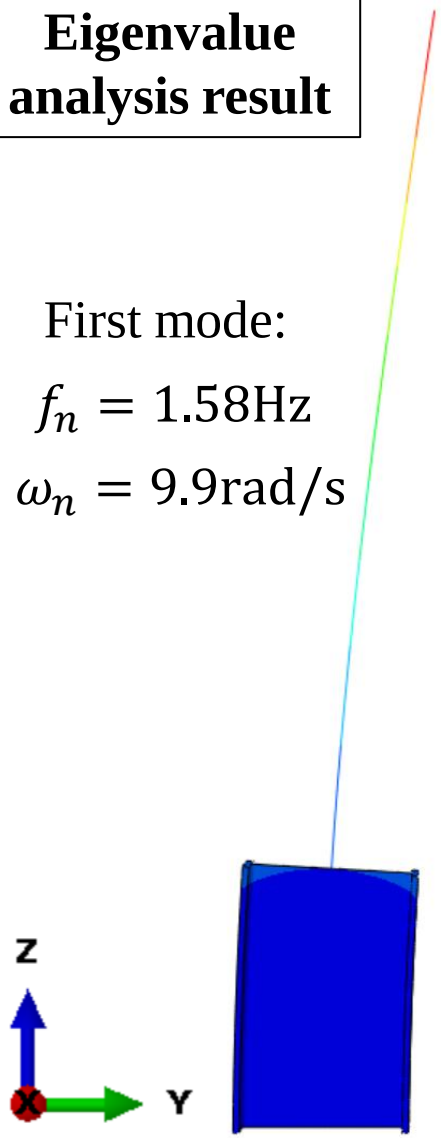


Eigenvalue analysis result

First mode:

$$f_n = 1.58\text{Hz}$$

$$\omega_n = 9.9\text{rad/s}$$



Hand calculation

$$k = \frac{3EI_x}{l^3} = 10500N/mm$$

$$= 1.05 \times 10^7 \text{N/m}$$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = 1.61\text{Hz}$$

Similar to the 1.58 Hz obtained numerically

2. Modifications of the UVC model for considering stiffness-proportional damping

2. Modified UVC model – modification of stress output

I modified the UVC subroutine so it can take an additional input parameter, β_R, to consider Rayleigh stiffness-proportional damping in UMAT. The modification follows the instruction on ABAQUS manual (ABAQUS, 2019) – once the element stress has been computed from the constitutive law, an additional damping-related stress component is added,

$$\sigma_m = \sigma + \beta_R D^{el} \ddot{\varepsilon}$$

σ_m – updated stress output. This modified stress then enters the Abaqus force equilibrium check at each increment;

σ – stress calculated from the constitutive law of UVC model at each increment;

β_R – Rayleigh damping coefficient;

D^{el} – elastic stiffness matrix;

$\ddot{\varepsilon}$ – strain rate of the increment.

Modifications to the UVC subroutine in multiaxial case

```
strain_rate = dstran / dtime
stress = sigma + beta_r * MATMUL(elasticity_matrix, strain_rate)
```

2. Modified UVC model – modification of tangent modulus

After the subroutine modification in the previous slide, I found this implementation is unstable. If the model is elastic, typically the analysis can converge; however, if the model material is elastoplastic, convergence cannot be guaranteed. I discussed with Diego on this matter a lot and checked the stress-strain outputs from many non-converging cases. Through this process, we identified the root cause of the problem.

As we use the stress with additional damping term, σ_m, for equilibrium check, we should use this stress's tangent modulus for strain increment update when force equilibrium is not reached, rather than using the tangent of original stress output, σ. This can achieve better convergence performance. The **modified tangent modulus** is,

$$D_m^{tg} = \frac{d\sigma_m}{d\varepsilon} = \frac{d\sigma}{d\varepsilon} + \beta_R D^{el} \frac{d\varepsilon}{d\varepsilon dt} = D^{tg} + \frac{\beta_R}{dt} D^{el}$$

D_m^{tg} – modified tangent modulus matrix based on the modified stress-strain relationship.

D^{tg} – original tangent modulus matrix based on the original constitutive stress-strain relationship;

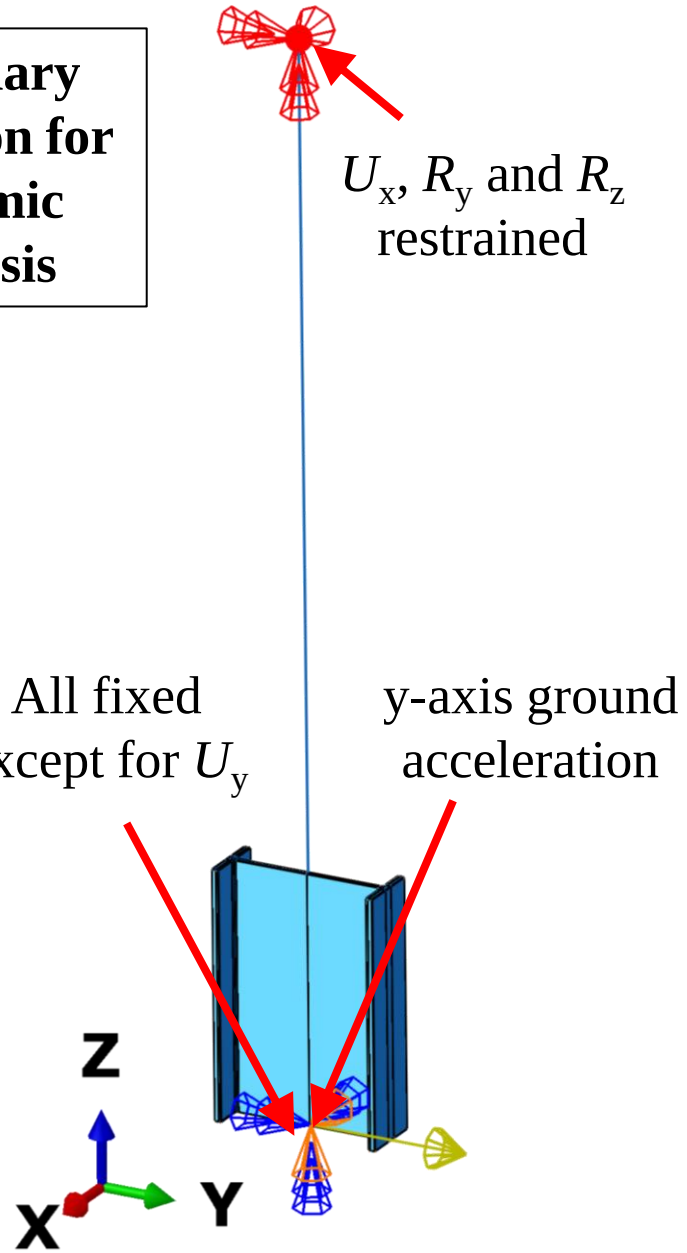
Modifications to the UVC subroutine in multiaxial case

```
ddsdde = ddsdde + beta_r * elasticity_matrix / dtime
```

3. Elastic SDF system dynamic analysis

3. Elastic SDF dynamic analysis – model details

Boundary condition for dynamic analysis



According to structural dynamics (Chopra 2012), when a SDF system is subject to a harmonic ground motion:

$$\ddot{u}_g(t) = \ddot{u}_{go} \sin \omega t$$

The maximum acceleration of the mass under steady state should be *TR* times of the ground motion amplitude:

$$\mathrm{TR} = \frac{\ddot{u}_o^t}{\ddot{u}_{go}} = \left\{ \frac{1 + [2\zeta(\omega/\omega_n)]^2}{\left[1 - (\omega/\omega_n)^2\right]^2 + [2\zeta(\omega/\omega_n)]^2} \right\}^{1/2}$$

In this test, ground motion frequency ω, is selected to be 9.9 rad/s, which is similar to structural frequency ω_n, to create resonance. Ground motion amplitude is selected to be 1 m/s^2.

In this case, if under a damping ratio of 2.5%, *TR* = 20.0 theoretically.

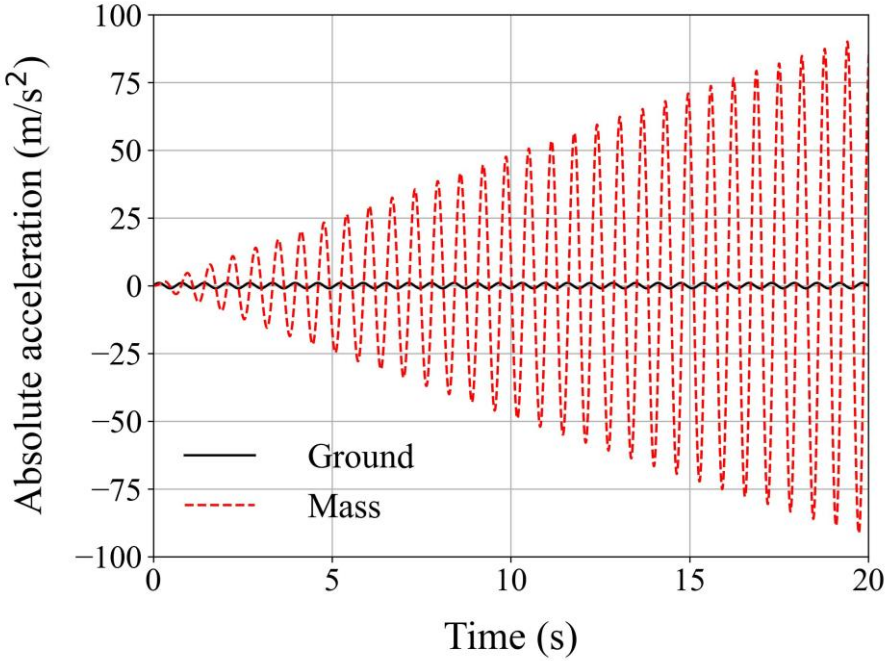
3. Elastic SDF dynamic analysis – validation of the built-in elastic material system

Then, I first validated the SDF FE model with built-in elastic material against theoretical solution. Two analyses were conducted for comparison – one without damping and the other with 2.5% damping ratio. The damped model is damped solely through stiffness-proportional damping coefficient, which can be calculated by,

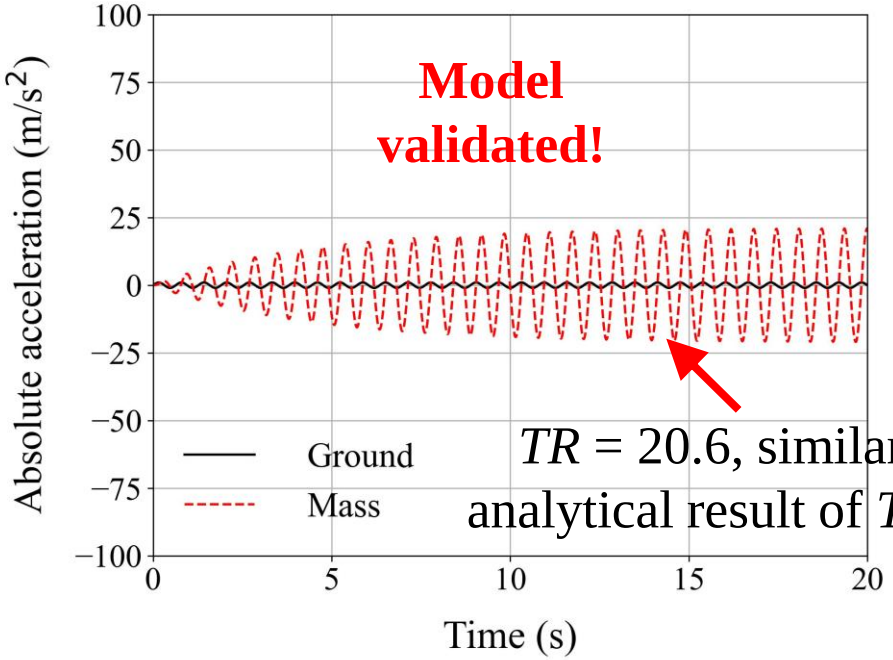
$$C = \alpha M + \beta K \rightarrow \xi_n = \frac{1}{2\omega_n}\alpha + \frac{\omega_n}{2}\beta, \alpha = 0 \rightarrow \beta = \omega_n / 2\xi_n = \frac{9.9}{2 \times 0.025} = 0.005$$

With built-in elastic material, under the conditions of β = 0 and β =0.005, the following results were obtained,

No damping, β = 0



2.5% damping ratio, β = 0.005

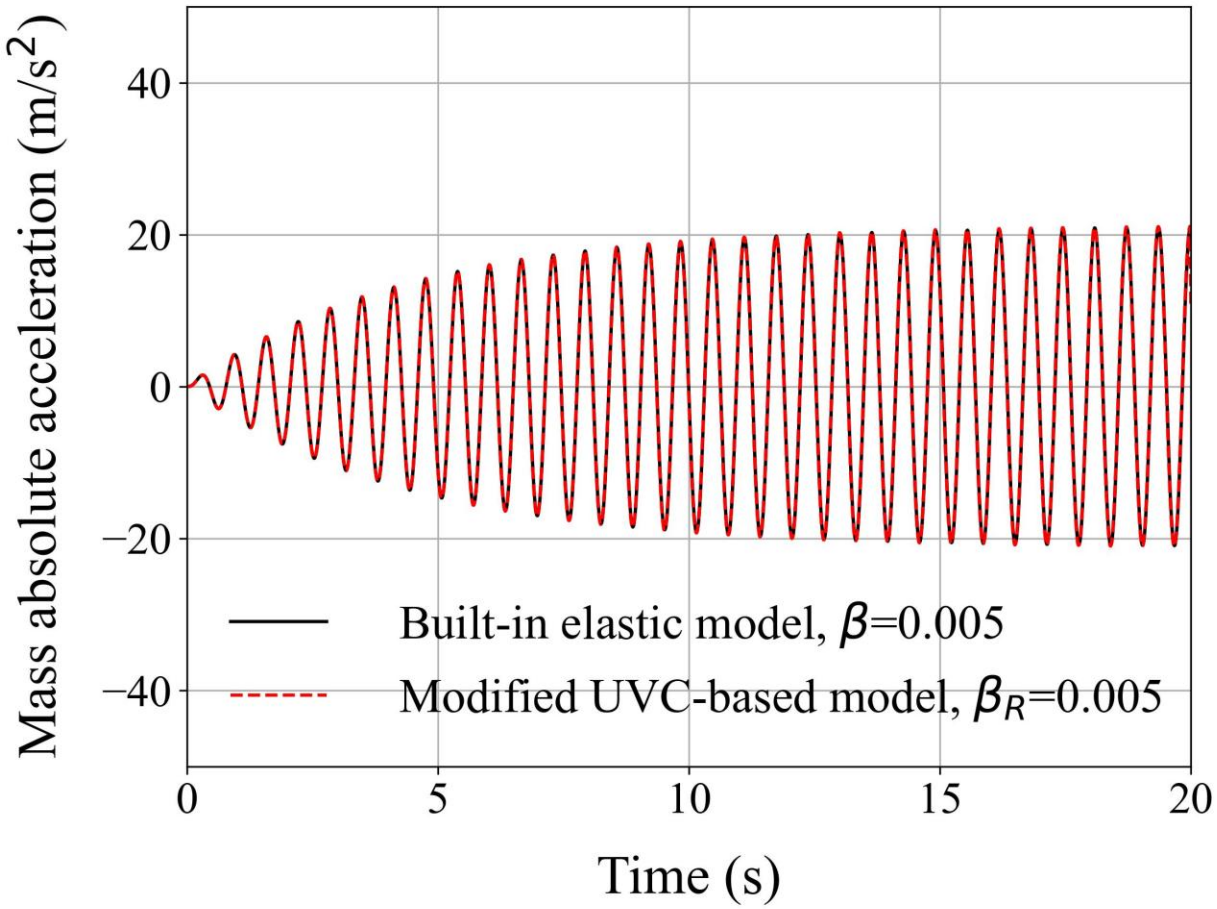


TR = 20.6, similar to the analytical result of *TR* = 20.0

3. Elastic SDF dynamic analysis – modified UVC-based elastic system

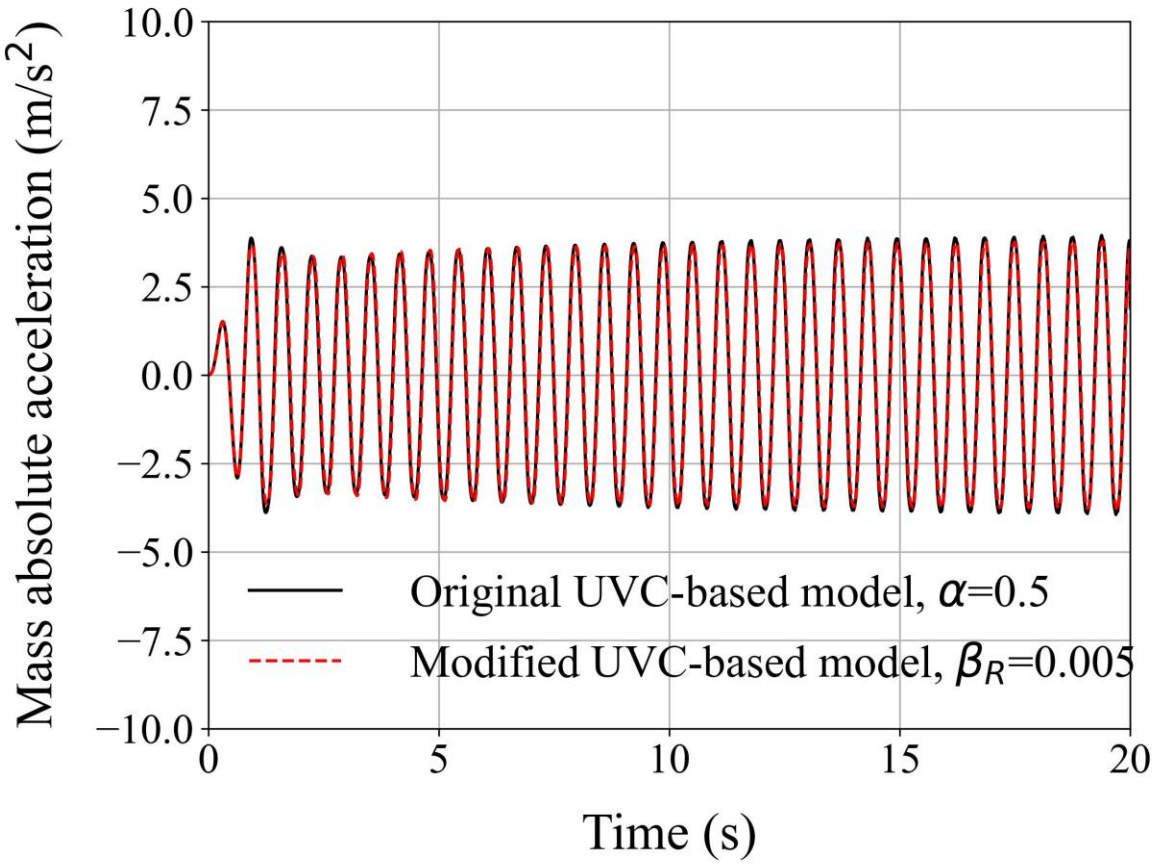
Then, I compared the dynamic simulation results of: 1) an SDF model with Abaqus built-in elastic material and β = 0.005, and 2) an SDF model with modified UVC-based elastic material and β_R = 0.005. Both models aim to achieve the same damping ratio of 2.5%.

Their mass absolute acceleration responses match good in the elastic case, as shown by the figure below.



4. Inelastic SDF system dynamic analysis

4. Inelastic SDF system dynamic analysis

For inelastic systems, as mentioned before, I compared the simulation results of, 1) an SDF model with original UVC model and built-in mass-proportional damping, and 2) an SDF model with modified UVC model. A 2.5% damping ratio is considered for both models – in this case, mass-proportional damping factor $\alpha = 0.5$ for the first model and $\beta_R = 0.005$ for the second model.

Their mass absolute acceleration responses match very good again, as shown by the figure below.